

COST OF CAPITAL

(pricing risk, CAPM, effect of leverage, estimating required rates)

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(PRICING RISK, CAPM, EFFECT OF LEVERAGE, ESTIMATING REQUIRED RATES)

Source

- *Corporate Finance*, Jonathan B. Berk and Peter DeMarzo: Ch 10, 11, 12
- *Principles of Corporate Finance*, Brealey, Myers, Allen, Edmans (14th Ed.): Ch 7
- *Investments*, Bodie, Kane, Marcus (12th Ed.): Ch 9

Capital markets and the pricing of risk

Risk and return

- Return is used to compare different securities. The **return** indicates the percentage increase in the value of an investment per dollar initially invested in the security. Each possible return has some likelihood, which can be summarized by a *probability distribution*. Given the probability distribution of returns, we can compute the expected return (weighted average of possible returns):

Expected (Mean) Return

$$\text{Expected Return} = E[R] = \sum_R p_R \times R$$

- It is the return we would earn on average if we could repeat the investment many times, drawing the return from the same distribution each time.
- Two common measures of the risk of a probability distribution are its **variance** and **standard deviation (volatility)**:

Variance and Standard Deviation of the Return Distribution

$$Var(R) = E[(R - E[R])^2] = \sum_R p_R \times (R - E[R])^2$$

$$SD(R) = \sqrt{Var(R)}$$

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$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + \mathbb{E}[X]^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + \mathbb{E}[X]^2 \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2\end{aligned}$$

- The variance measures the variability of returns, i.e. how “spread out” the distribution of the return is. When the return is risk-free, the variance is **0**.
- While *variance and standard deviation are the most common measures of risk*, they do not differentiate upside and downside risk. Alternative measures that focus on *downside risk* include the **semi-variance** (variance of the losses only) and the **expected tail loss** (the expected loss in the worst x% of outcomes).
- *In real life, in most situations we do not know the explicit probability distribution*, o/w we could compute their expected returns and volatilities. Without that information, a popular approach is to *extrapolate from historical data* (believing that the distribution of future returns should mirror that of past returns).
- **Realized return** is the return that actually occurs over a particular time period:

$$\begin{aligned}R_{t+1} &= \frac{\text{Div}_{t+1} + P_{t+1}}{P_t} - 1 = \frac{\text{Div}_{t+1}}{P_t} + \frac{P_{t+1} - P_t}{P_t} \\ &= \text{Dividend Yield} + \text{Capital Gain Rate}\end{aligned}$$

- The **average annual return** of an investment during some historical period is simply the average of the realized returns for each year. *If the probability distribution of the returns is the same over time*, the average return provides an estimate of the expected return. Since we do not know the true mean, the best estimate of the mean is the average annual return.

Average Annual Return of a Security

$$\bar{R} = \frac{1}{T}(R_1 + R_2 + \cdots + R_T) = \frac{1}{T} \sum_{t=1}^T R_t$$

- To quantify this difference in variability from one investment to the other, we can estimate the standard deviation of the probability distribution.

Variance Estimate Using Realized Returns

$$\text{Var}(R) = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2$$

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- If the future looked like the past, we could *assume that investors will require the same return in the future*. However,
 - o we do not know what investors expected in the past; we can only observe the actual returns that were realized
 - o the average return is just an estimate of the true expected return, and is subject to *estimation error*
- We can measure this estimation error of a statistical estimate by its **standard error**. The standard error is the standard deviation of the estimated value of the mean of the actual distribution around its true value. If the distribution of returns is IID (independent and identically distributed: the likelihood that the return has a given outcome is the same each year and does not depend on past returns) each year, then:

Standard Error of the Estimate of the Expected Return

$$SD \text{ (Average of Independent, Identical Risks)} = \frac{SD \text{ (Individual Risk)}}{\sqrt{\text{Number of Observations}}}$$

- The avg. return will be within 2 (more accurately 1.96) standard errors of the true expected return approximately 95% of the time (*if returns are IID and normally distributed*), so we can estimate the true expected value using a 95% confidence interval:

$$\text{Historical Average Return} \pm (2 \times \text{Standard Error})$$

- **The problem with this approach:** individual stocks are even more volatile than portfolios and may have been in existence for only few years, thus not having enough data to have a good estimate about the true expected return, because the estimation error will be large. Thus, *the past avg return is not a reliable expectation for the future*.

Trade-off between risk and return

- **Risk aversion:** investors would not choose to hold a portfolio that is more volatile unless they expected to earn a higher return
- **Excess return / risk premium:** it is the difference between the average return for the investment and the average return for a risk-free investment (Treasury bills)

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Pricing Risk

- There is a positive relationship between risk and return (in case of portfolios), in the sense that investments with higher volatility (risk) have rewarded investors with higher average returns → *proof for risk aversion*: riskier investments must offer investors higher average returns to compensate them for the extra risk they are taking on

Diversification

- Usually, stock prices and dividends fluctuate due to two types of news: *firm-specific* news and *market-wide* news. Thus, two types of risk can be differentiated:
 - **Firm-specific, idiosyncratic, unique, or diversifiable risk:**
Fluctuations of a stock's return that are due to firm-specific news are *independent risks*.
 - **Systematic, undiversifiable, or market risk**
Fluctuations of a stock's return that are due to market-wide news represent *common risk*.
- In a large portfolio of individual stocks, firm-specific risk can be diversified. The systematic risk, however, will affect all firms and cannot be diversified. A portfolio which consists of firms that are affected only by idiosyncratic risk should earn the risk free rate (*law of one price*). This no-arbitrage argument states that: **the risk premium for diversifiable risk is zero, so investors are not compensated for holding firm-specific risk**. However, since investors are risk averse, they will demand a risk premium to hold systematic risk; o/w they would be better off selling their stocks and investing in risk-free bonds. To conclude: **the risk premium of a security is determined by its systematic risk and does not depend on its diversifiable risk**.
- Two common measures of the risk of a probability distribution are its **variance** and **standard deviation (volatility)**. While *variance and standard deviation are the most common measures of risk*, they do not differentiate upside and downside risk.
- **We cannot use SD to determine the risk premium** investors will earn, since SD is a measure of *total risk* (both firm-specific and systematic) → *no relationship between risk*

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and avg return for *individual stocks*. However, it is still a reasonable measure of risk for well-diversified portfolios.

Measuring systematic risk

- We would like to know how sensitive the stock is to systematic shocks that affect the economy as a whole, i.e. how much of the variability of its return is due to systematic, market-wide risks versus diversifiable, firm-specific risks. For this we can look at the average change in its return for each 1% change in the return of *a portfolio that fluctuates solely due to systematic risk*.
- The **efficient portfolio** contains only systematic risk. This portfolio cannot be diversified further, i.e. there is *no way to reduce the risk of the portfolio without lowering its expected return*. A natural candidate for an efficient portfolio is the **market portfolio** - *a portfolio of all stocks and securities traded in the capital markets* (ex. S&P 500). By assuming that the market portfolio is efficient, we can measure the systematic risk of a security by calculating the sensitivity of the security's return to the return of the market portfolio, known as the β of the security. **The beta of a security is the expected % change in its (excess) return given a 1% change in the (excess) return of the market portfolio.** The market beta is 1, stocks in cyclical industries are likely to be more sensitive to systematic risk and have betas that exceed 1, whereas stocks of non-cyclical firms tend to have betas that are less than 1.
- If a stock had **negative beta**, it would mean that it has a negative risk premium, i.e. it would have an expected return below the risk-free rate. Owning such an asset will provide *insurance* (investors accept less return in exchange for insurance) against the systematic risk of other stocks in the portfolio when times are bad.
- Thus, the beta of a security is the appropriate measure of its risk because *beta is proportional to the risk the security contributes to the optimal risky portfolio*.

Cost of capital and beta

- Firms use cost of capital to analyze whether to proceed with a project. **Cost of capital** is the *expected return available on alternative investments in the market with similar risk and term*. For risky investments the cost of capital is $r_f + \text{risk premium}$. As long as the cost of

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capital is below the rate of return the company earns by using its capital, it's a good investment.

- The size of the *risk premium* depends on investors' risk aversion (we can measure risk aversion indirectly through the risk premium they demand). The market risk premium is the reward investors expect to earn for holding a portfolio with a beta of 1 - the market portfolio itself.

$$\text{Market Risk Premium} = E[R_{Mkt}] - r_f$$

- We can use the beta of the investment to determine the scale of the investment in the market portfolio that has equivalent systematic risk. The most important method for estimating the cost of capital in practice is the **CAPM**:

Estimating the Cost of Capital of an Investment from Its Beta

$$\begin{aligned} r_I &= \text{Risk-Free Interest Rate} + \beta_I \times \text{Market Risk Premium} \\ &= r_f + \beta_I \times (E[R_{Mkt}] - r_f) \end{aligned}$$

Optimal portfolio and CAPM

Portfolio risk and return

- The (expected) return of a portfolio is the weighted average of the (expected) returns of the stocks in the portfolio.

$$E[R_p] = E\left[\sum_i x_i R_i\right] = \sum_i E[x_i R_i] = \sum_i x_i E[R_i]$$

- To determine the risk of a portfolio, the return and risk of the stocks is not enough, we need the *covariance* (and *correlation*) of them: the degree to which the stocks face common risks, and their returns move together.
- **Covariance** is the expected product of the deviations of two returns from their means.

Covariance between Returns R_i and R_j

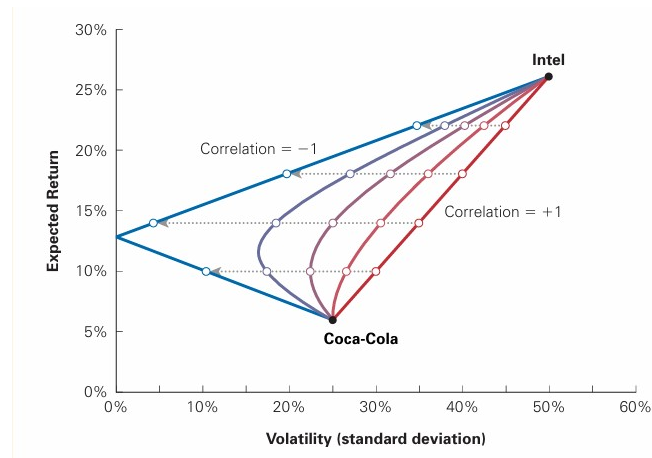
$$\text{Cov}(R_i, R_j) = E[(R_i - E[R_i])(R_j - E[R_j])]$$

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- **Correlation** controls for the volatility of each stock and quantifies the strength of the relationship between them. *Correlation has no effect on the return of a portfolio, but it has on its volatility.* With two perfectly negatively correlated stocks the risk would be **0**.

$$\text{Corr}(R_i, R_j) = \frac{\text{Cov}(R_i, R_j)}{SD(R_i) SD(R_j)}$$



- The variance of a portfolio depends on the variance of the individual stocks and on the covariance between them:

The Variance of a Two-Stock Portfolio

$$\text{Var}(R_p) = x_1^2 \text{Var}(R_1) + x_2^2 \text{Var}(R_2) + 2x_1x_2 \text{Cov}(R_1, R_2)$$

- Almost all of the benefits of diversification can be achieved with 20-30 stocks.

Volatility of a Portfolio with Arbitrary Weights

$$SD(R_p) = \sum_i \overbrace{x_i \times SD(R_i) \times \text{Corr}(R_i, R_p)}^{\text{Security } i \text{'s contribution to the volatility of the portfolio}}$$

$\uparrow$
Amount
of i held

$\uparrow$
Total
risk of i

$\uparrow$
Fraction of i 's
risk that is
common to P

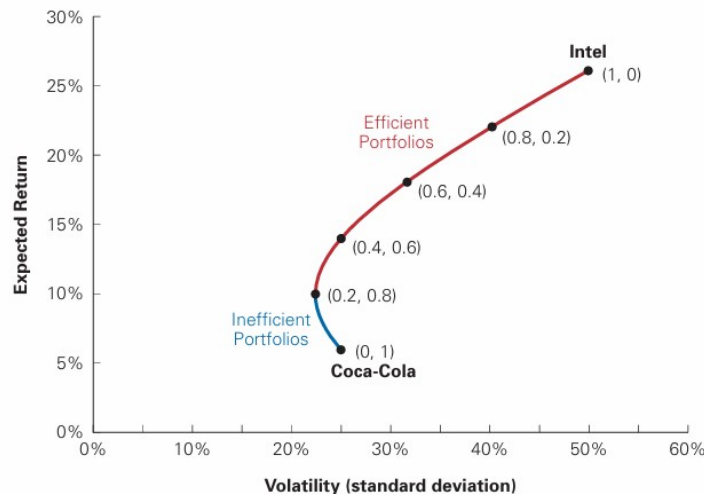
- The expected return of a portfolio is equal to the weighted average expected return, but the volatility of a portfolio is less than the weighted average volatility: we can eliminate some volatility by diversifying.

Efficient portfolio

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- How to create an efficient portfolio? In case of two stocks, given the volatility and return of them, calculate the risk and return of the portfolio. Then the different combination of weights of the two stocks can be plotted on a risk-return graph. When the exp. return is the highest and the volatility the lowest, that's the weight combination you want to choose. Actually, all the portfolios that lie on the **red** line (above the global minimum variance portfolio) connecting these weight-combinations are *efficient portfolios*. A portfolio is *inefficient* whenever it is possible to find another portfolio that is better in terms of both expected return and volatility. Intuitively, efficient portfolios are those when there is no other portfolio of the two stocks that offers a higher expected return with lower volatility. The upper part of the curve, the red curve, is the **efficient frontier**. Adding more stocks to the portfolio improves the efficient frontier by allowing greater diversification. All investors will choose from this red curve based on their risk-return preferences.



- We could keep some of our money also in risk-free assets, but this will reduce the expected return. Contrary, we could borrow, to invest even more in the stock market. The **optimal portfolio** will consist of an amount invested in the risk-free security versus the risky ones. The expected return and volatility would be ($1-x$ weight is put in the risk-free asset):

$$\begin{aligned} E[R_{xP}] &= (1-x)r_f + xE[R_P] \\ &= r_f + x(E[R_P] - r_f) \end{aligned}$$

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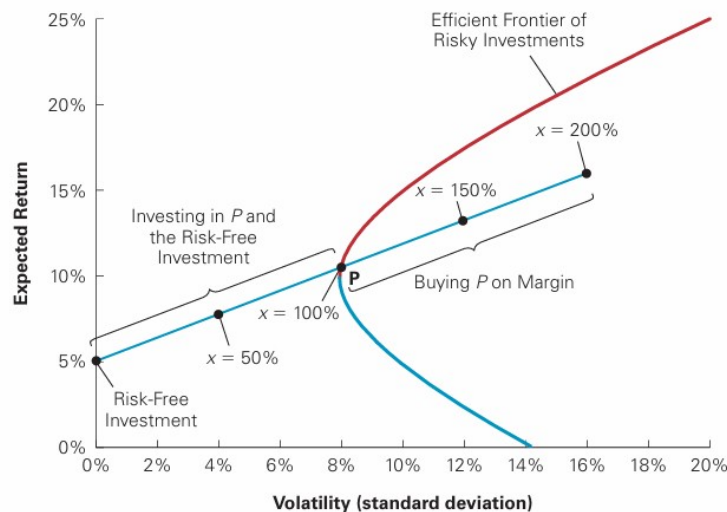
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$$\begin{aligned}
 SD(R_{xP}) &= \sqrt{(1-x)^2 \text{Var}(r_f) + x^2 \text{Var}(R_P) + 2(1-x)x \text{Cov}(r_f, R_P)} \\
 &= \sqrt{x^2 \text{Var}(R_P)} \\
 &= xSD(R_P)
 \end{aligned}$$

↙ ↘
0

- The blue line on the below chart illustrates combinations of volatility and expected return for different choices of x (*weight in the portfolio*). As we increase the fraction x invested in P , we increase both our risk and our risk premium proportionally. If we go beyond P , we are short selling the risk-free investment, so we must pay the risk-free return; in other words, we are borrowing money at the risk-free interest rate. *Margin investing* can provide higher expected returns than investing in P using only the funds we have available, but the risk is higher than that of portfolio P .
- P is not the best portfolio to combine with the r_f asset, because the line can be steeper. The slope of the line through a given portfolio P is often referred to as the **Sharpe ratio** (*reward-to-volatility ratio*) of the portfolio, which shows the expected return for any level of volatility. The higher the ratio, the better (the steeper the line).

$$\text{Sharpe Ratio} = \frac{\text{Portfolio Excess Return}}{\text{Portfolio Volatility}} = \frac{E[R_P] - r_f}{SD(R_P)}$$



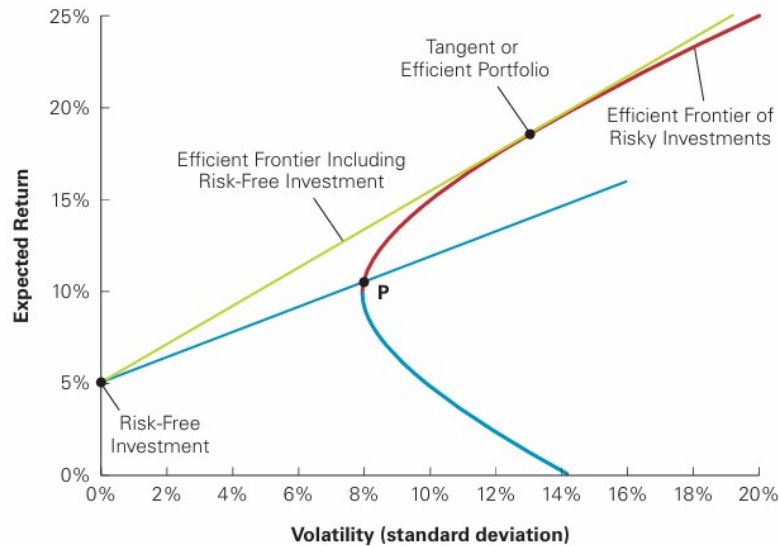
- The optimal portfolio to combine with the risk-free asset will be the one with the highest *Sharpe ratio*, where the line with the risk-free investment just touches, and so is tangent to, the efficient frontier of risky investments. The portfolio that generates this tangent line is

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known as the **tangent portfolio**. The tangent portfolio provides the biggest reward per unit of volatility of any portfolio available.

- The combinations of the risk-free asset and the tangent portfolio provide the *best risk and return trade-off* available to an investor.



- **Every investor should invest in the tangent portfolio independent of his or her taste of risk** but will choose different weight for the tangent portfolio and risk-free asset.

CAPM

- The Capital Asset Pricing Model is a set of predictions concerning equilibrium expected returns on risky assets. The basic idea of each variant of CAPM is that *risk premium will be proportional to the exposure to systematic risk and independent of firm-specific risk*. CAPM is a vital tool for linking risk to expected return.
- Assumptions (more complex list in the table):
 - o No taxes and no transaction costs.
 - o Possibility to borrow/lend at r_f .
 - o Investors have *homogeneous expectations* (all investors face the same publicly available information, so their estimates should be the same) regarding the volatilities, correlations, and expected returns of securities (same input list).
 - o Investors hold only efficient portfolios of traded securities.

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Table 9.1

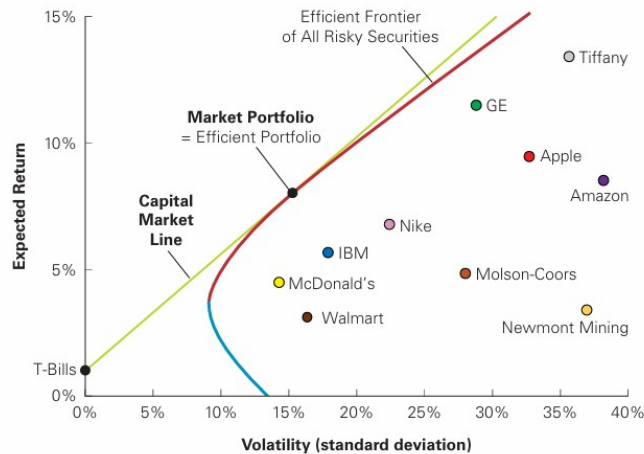
The assumptions of the CAPM

- | |
|---|
| 1. Individual behavior |
| a. Investors are rational, mean-variance optimizers. |
| b. Their common planning horizon is a single period. |
| c. Investors all use identical input lists, an assumption often termed homogeneous expectations . Homogeneous expectations are consistent with the assumption that all relevant information is publicly available. |
| 2. Market structure |
| a. All assets are publicly held and trade on public exchanges. |
| b. Investors can borrow or lend at a common risk-free rate, and they can take short positions on traded securities. |
| c. No taxes. |
| d. No transaction costs. |

- To identify the efficient portfolio, we must know the expected returns, volatilities, and correlations between investments. Investors rather observe security prices and bid those prices up and down. The exp. rate of return can at most be inferred from the prices investors pay compared to the cash flows those investments are expected to generate. It is hard to forecast these, but the CAPM does not require any knowledge about the expected return of each security.
- The CAPM assumes that all investors optimize their portfolios using the Markowitz model of efficient diversification. *The efficient or tangent portfolio of risky securities (the portfolio that all investors hold) must equal the market portfolio.* It is the same for everyone, since each investor uses the same input list (same estimates for exp. return, variance, covariance). When the tangent line goes through the market portfolio, it is called the **capital market line (CML)**.
- The market portfolio includes all assets. Suppose that a company is not included in the market portfolio. Then no one would buy it, and its price would fall until it would be attractive to include it in the portfolio. So, in the end, this price adjustment guarantees that each stock will be included in the market portfolio, which is also the optimal tangency portfolio in the CAPM world.

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- If we don't know the expected return of a security or the cost of capital of an investment, we can use the CAPM to find it by using the market portfolio as a benchmark.

The CAPM Equation for the Expected Return

$$E[R_i] = r_i = r_f + \underbrace{\beta_i \times (E[R_{Mkt}] - r_f)}_{\text{Risk premium for security } i}$$

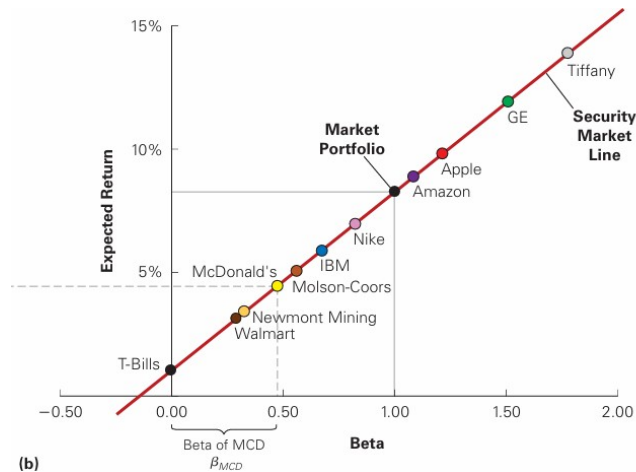
- The r_f compensates investors for “waiting” (time value of money), the risk premium compensates for “worrying” (about investment returns). *AGAIN: investors are not compensated for firm-specific risk!*

$$\beta_i = \frac{\overbrace{SD(R_i) \times \text{Corr}(R_i, R_{Mkt})}^{\text{Volatility of } i \text{ that is common with the market}}}{SD(R_{Mkt})} = \frac{\text{Cov}(R_i, R_{Mkt})}{\text{Var}(R_{Mkt})}$$

- Because investors can eliminate firm-specific risk by diversifying their portfolios, *the right measure of risk is the investment's beta* with the market portfolio, β_i .
- **Security Market Line (SML)** is the line along which all **individual securities** *should* lie when plotted according to their expected return and beta. Because the market's beta is 1, the slope is the risk premium of the market portfolio. The security market line provides a benchmark for the evaluation of investment performance. “Fairly priced” assets should plot on the SML. If a stock is perceived to be a good buy, or underpriced, it will provide an *expected return in excess of the fair return* stipulated by the SML. **Underpriced** stocks therefore plot **above** the SML: given their betas, their expected returns are greater than dictated by the CAPM. **Overpriced** stocks plot **below** the SML. The difference between the fair and actually expected rate of return on a stock is called the stock's **alpha, α** .

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- Alpha is part of the **index model**. The index model states that the realized excess return on any stock is the sum of the realized excess return due to marketwide factors $\beta_i r_M$, a non-market premium, α_i , and firm-specific outcomes summarized by e_i (it is eliminated when taking its expectation).

$$R_i = \alpha_i + \beta_i R_M + e_i$$

$$E(R_i) = \alpha_i + \beta_i E(R_M)$$

- The prediction of the CAPM is that for every stock, the equilibrium value of α_i is 0. The logic of the CAPM is that the only reason for a stock to provide a premium over the risk-free rate is that the stock imposes systematic risk for which the investor must be compensated. A positive alpha implies reward without risk. If there were a positive (negative) alpha stock, investors would buy (sell) it and drive the price up (down), so exp. return will decrease (increase). This *portfolio rebalancing* will continue until all values are driven to zero.
- The beta of a portfolio is the weighted average beta of the securities in the portfolio.

$$\begin{aligned} \beta_p &= \frac{Cov(R_p, R_{Mkt})}{Var(R_{Mkt})} = \frac{Cov(\sum_i x_i R_i, R_{Mkt})}{Var(R_{Mkt})} = \sum_i x_i \frac{Cov(R_i, R_{Mkt})}{Var(R_{Mkt})} \\ &= \sum_i x_i \beta_i \end{aligned}$$

Estimating required rates

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- We can add more stocks to the portfolio if their expected return (from CAPM) exceeds their required return. We can do this by selling some of our risk-free assets and investing the proceeds in asset i , for example. The **required return** is the *expected return that is necessary to compensate for the risk investment i will contribute to the portfolio*:

$$r_i \equiv r_f + \beta_i^P \times (E[R_P] - r_f)$$

$$\beta_i^P \equiv \frac{SD(R_i) \times \text{Corr}(R_i, R_P)}{SD(R_P)}$$

- Thus, our exp. return will increase (by $E[r_i] - r_f$), but we will also add the risk that asset i has in common with our portfolio ($SD(r_i) * \text{Corr}(r_i, r_P)$). Increasing the amount invested in i will increase the Sharpe ratio of portfolio P if its expected return $E[r_i]$ exceeds its required return given portfolio P . If i 's expected return exceeds its required return, then adding more of it will improve the performance of the portfolio.
- The final portfolio will be efficient if the *expected return of every security in it equals its required return* (we will add them until the exp. return won't be higher than the required return anymore).

Expected Return of a Security

$$E[R_i] = r_i \equiv r_f + \beta_i^{eff} \times (E[R_{eff}] - r_f)$$

- This states that *we can determine the appropriate risk premium for an investment from its beta with the efficient portfolio*. The efficient or tangent portfolio, which has the best possible Sharpe ratio of any portfolio in the market, provides the benchmark that identifies the systematic risk present in the economy.
- To understand required rates better, check the below example:

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Problem

You are currently invested in the Omega Fund, a broad-based fund with an expected return of 15% and a volatility of 20%, as well as in risk-free Treasuries paying 3%. Your broker suggests that you add a real estate fund to your portfolio. The real estate fund has an expected return of 9%, a volatility of 35%, and a correlation of 0.10 with the Omega Fund. Will adding the real estate fund improve your portfolio?

Solution

Let R_{re} be the return of the real estate fund and R_O be the return of the Omega Fund. From Eq. 11.19, the beta of the real estate fund with the Omega Fund is

$$\beta_{re}^O = \frac{SD(R_{re}) \text{Corr}(R_{re}, R_O)}{SD(R_O)} = \frac{35\% \times 0.10}{20\%} = 0.175$$

We can then use Eq. 11.20 to determine the required return that makes the real estate fund an attractive addition to our portfolio:

$$r_{re} = r_f + \beta_{re}^O (E[R_O] - r_f) = 3\% + 0.175 \times (15\% - 3\%) = 5.1\%$$

Because its expected return of 9% exceeds the required return of 5.1%, investing some amount in the real estate fund will improve our portfolio's Sharpe ratio.

Estimating the cost of capital

Equity cost of capital

- The **cost of capital** is the best expected return available in the market on investments with similar risk. Using CAPM we can identify investments with similar risk. The cost of capital of any investment opportunity equals the expected return of available investments with the same beta. Investors will require a risk premium comparable to what they would earn by taking the same market risk through an investment in the market portfolio. Thus, for a firm considering a new project, the CAPM can provide the *required rate of return* that the project needs to yield, based on its beta, to be acceptable to investors.

The CAPM Equation for the Cost of Capital (Security Market Line)

$$r_i = r_f + \underbrace{\beta_i \times (E[R_{Mkt}] - r_f)}_{\text{Risk premium for security } i}$$

- To value a stock, we need to calculate the **equity cost of capital** using CAPM. The **market risk premium** is needed for this, which can be estimated using **historical average excess return** of the market over the risk-free interest rate. However, this has two drawbacks.

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Even when using data of several decades, the SE (standard error) of the estimates is large. Moreover, they are backward looking, so may not represent the current expectations about the future. Another approach would be the **fundamental approach**:

$$r_{Mkt} = \frac{Div_1}{P_0} + g = \text{Dividend Yield} + \text{Expected Dividend Growth Rate}$$

- **Beta** is the appropriate measure of risk for a well-diversified portfolio. It can be estimated based on the **stock's historical sensitivity** if its beta is relatively stable over time. Usually, it is done by estimating correlations and volatilities *from two to five years of weekly or monthly returns* and using the S&P 500 as the market portfolio. It can also be calculated by plotting the stock's and the market's excess returns and drawing the **best-fitting line** through these points. The slope of this line is the β . The best-fitting line is identified through linear regression.

$$(R_i - r_f) = \alpha_i + \beta_i (R_{Mkt} - r_f) + \varepsilon_i$$

$$E[R_i] = \underbrace{r_f + \beta_i (E[R_{Mkt}] - r_f)}_{\text{Expected return for } i \text{ from the SML}} + \underbrace{\alpha_i}_{\text{Distance above/below the SML}}$$

- The CAPM can provide much more accurate estimates of expected returns for stocks than we could obtain from their historical average return.

Debt cost of capital

- **Debt cost of capital** is the cost of capital that a firm must pay on its debt. It is the expected return required by the creditors of a firm.
- The **yield to maturity** (YTM) of a bond is the IRR an investor will earn from holding the bond to maturity and receiving its promised payments. When the PD (probability of default) is low, YTM is a good estimate of expected return, if PD is higher, YTM will overstate it. Based on credit ratings, annual default rates can be found on the internet. With a PD p and expected loss L :

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$$r_d = (1 - p)y + p(y - L) = y - pL$$
$$= \text{Yield to Maturity} - \text{Prob}(\text{default}) \times \text{Expected Loss Rate}$$

- Alternatively, it can also be calculated using **CAPM**. Beta estimates for debt instruments based on historical data would not be reliable since bank loans and many corporate bonds are not traded frequently. But we could estimate them using betas of bond indices with same rating category.

Company and Project cost of capital

- The **company cost of capital** is the opportunity cost of capital for an investment in all the assets of the firm and is the correct discount rate for its average risk projects. There are two ways to estimate the company cost of capital: estimate the company's asset beta and use the CAPM to calculate the cost of capital or calculate the weighted average of the returns demanded by equity and debt investors. The second method is much easier, since we know the actual cost of debt (cost of borrowing), but we do not observe the debt beta directly and it is hard to estimate.
- *Projects should be evaluated on their own*, separate from any financing decisions. It is assumed that the project is all-equity financed. Because a new project is not itself a publicly traded security, we cannot estimate its cost of capital and beta based on historical data. We should *identify comparable firms* in the same line of business as the project we are considering undertaking (sometimes the only such company is the one undertaking the project) and use their beta estimates.
- If a company is **all-equity financed**, then by owning its stock we own a portfolio of its underlying assets, thus if the firm's average investment has similar market risk to the project, then we can use its equity beta and cost of capital as estimates for beta and the cost of capital of the project.
- It is complicated when a company is **levered**, bc the cash flows generated by the firm's assets are used to pay both debt and equity holders. We can recreate a claim on the firm's assets by holding both its debt and equity simultaneously. The return of the firm's assets is therefore the same as the return of a portfolio of the firm's debt and equity combined.

COST OF CAPITAL

(pricing risk, CAPM, effect of leverage, estimating required rates)

- The **asset cost of capital / unlevered cost of capital** is the expected return required by the firm's investors to hold the firm's underlying assets (it is the average rate the company makes on its projects). This is the return that can be distributed to both equity and debt holders.

Asset or Unlevered Cost of Capital

$$r_U = \frac{E}{E+D}r_E + \frac{D}{E+D}r_D$$

Asset or Unlevered Beta

$$\beta_U = \frac{E}{E+D}\beta_E + \frac{D}{E+D}\beta_D$$

- Because **cash** represents a risk-free asset on the firm's balance sheet, and reduces the average risk of the firm's assets, we are interested in the *risk of the firm's underlying business operations*, the firm's **enterprise value** (equity + net debt).

$$\text{Net Debt} = \text{Debt} - \text{Excess Cash and Short-Term Investments}$$

- When a firm finances its own project using debt, it will benefit from tax reduction, so we should use the **after-tax WACC**:

Weighted Average Cost of Capital (WACC)

$$r_{wacc} = \frac{E}{E+D}r_E + \frac{D}{E+D}r_D(1 - \tau_C)$$

- WACC is based on the effective after-tax cost of debt, whereas the unlevered cost of capital (*pretax WACC*) is based on the firm's pretax cost of debt.

WACC summary:

- The **unlevered cost of capital** (or pretax WACC) is the expected return investors will earn holding the firm's assets. In a world with taxes, it can be used to evaluate an all-equity financed project with the same risk as the firm.
- The **weighted average cost of capital** (or WACC) is the effective after-tax cost of capital to the firm. Because interest expense is tax deductible, the *WACC is less than the expected return of the firm's assets*. In a world with taxes, the WACC can be used to evaluate a project with the same risk and the same financing as the firm itself.

COST OF CAPITAL

(pricing risk, CAPM, effect of leverage, estimating required rates)

$$r_{wacc} = r_U - \frac{D}{E + D} \tau_C r_D$$